

Products

Lawvere–Schanuel’s first exercise, answered for
Kalkulus I (MBLK37E) and *Lineáris algebra I* (MBLK15E)

Abstract

Exercise 1 of Lawvere and Schanuel’s *Conceptual Mathematics* asks the reader to find products “in life” and in other circumstances. This note answers it three times over. Section 1 states the one test that decides whether something is a product. Section 2 runs the test on twelve everyday situations. Section 3 goes through the two *Tárgytematika* documents item by item and sorts every occurrence of the word *szorzat* into the two possible meanings — a *product object* (a pair) and a *multiplication map* (a map *out of* a pair). Section 4 does the same for the exercise sheets `kalkulus_gyakorlo.pdf`. Section 5 turns all of it into a protocol for a timed exam. Every mathematical claim made here is stated and proved in `RequestProject/ProductOverlap.lean`; the Lean name is printed in the margin of the claim, e.g. `type_isProduct`.

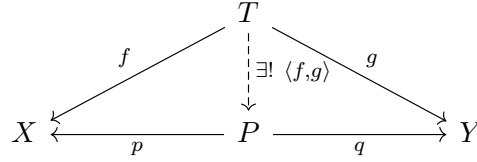
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1 The one test

Definition 1.1 (Lawvere–Schanuel). Let X and Y be objects of a category. An object P together with two maps $p : P \rightarrow X$ and $q : P \rightarrow Y$ is a *product* of X and Y if for every object T and every pair of maps $f : T \rightarrow X$, $g : T \rightarrow Y$ there is *exactly one* map $\langle f, g \rangle : T \rightarrow P$ with

$$\langle f, g \rangle; p = f, \quad \langle f, g \rangle; q = g.$$



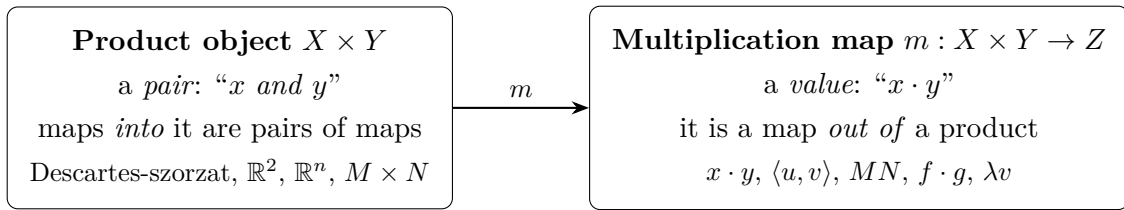
Lean: `ProductOverlap.IsProduct`.

The definition never mentions elements. It says: *to give one thing of type P is the same as to give one thing of type X and one thing of type Y* . That sentence is the exam-usable form of the test. Two consequences, both proved once and then reused:

- **Uniqueness.** Any two products of the same X, Y are isomorphic by a unique isomorphism commuting with the projections — `IsProduct.unique_up_to_iso`. So “the” product is legitimate language, and any construction passing the test is as good as any other.
- **Commutativity for free.** $(Y \times X, \text{snd}, \text{fst})$ also passes the test for X, Y , hence $X \times Y \cong Y \times X$ — `swap_isProduct`, `prod_comm_iso`. Choosing the colour first and the size second is the same as the other way round.

Product object versus multiplication map

This distinction organises everything below, and is the single most useful thing to carry into an exam.



Both are called *szorzat* in Hungarian and *product* in English, but they are different kinds of thing: the first is an object, the second an arrow. A pointwise product of functions is the composite of both:

$$f \cdot g = m \circ \langle f, g \rangle, \quad \mathbb{R} \xrightarrow{\langle f, g \rangle} \mathbb{R} \times \mathbb{R} \xrightarrow{m} \mathbb{R}.$$

Every theorem about products of functions in *Kalkulus I* is obtained by feeding a property of $\langle f, g \rangle$ (it is continuous/differentiable coordinatewise) into a property of m (it is continuous, and bilinear). See Sections 3 and 5.

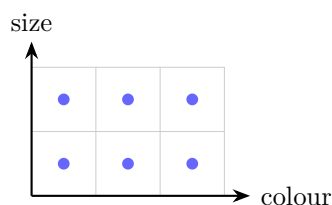
2 Products in life

Run the test of Definition 1.1 on each situation: *is specifying one thing of the whole exactly the same as specifying one thing of each part, with no leftover choice and no clash?*

Situation	The two projections	Why it passes (or fails)
A shirt in a shop	colour; size	Choosing a shirt = choosing a colour and a size. 3 colours, 2 sizes, 6 shirts. <code>shirts_card</code>
A square of a chessboard	file (a–h); rank (1–8)	“e4” is the pair; 64 = 8·8 squares. <code>card_prod_eq_mul</code>
An appointment	day; time of day	The calendar grid is literally drawn as a rectangle.

Situation	The two projections	Why it passes (or fails)
A point on the screen	x pixel; y pixel	The plane; the projections are the two rulers.
A colour on the screen	(R, G, B)	A triple product; 256^3 colours.
A person's ID record	name; date of birth	Passes only if any name may occur with any date — which is what “independent fields” means.
A meal deal	main course; dessert	$ \text{mains} \cdot \text{desserts} $ meals: the counting principle.
A PIN of two digits	first digit; second digit	$10 \cdot 10$; the projections are “read the first/second symbol”.
Two dice rolled together	red die; blue die	36 outcomes; independence of the two dice <i>is</i> the product structure.
A parallel pair of tasks	progress on task 1; on task 2	Passes if the tasks do not interact; the moment one blocks the other it is no longer a product.
A married couple's shopping list	his list; her list	Fails as stated: the joint list is a <i>union</i> , not a pair — a sum, not a product.
A journey Budapest–Szeged	departure city; arrival city	Fails: not every pair of cities is a legal journey. The set of journeys is a <i>subobject</i> of the product.

The last two rows matter as much as the others. “Number of possibilities multiplies” is the fingerprint of a product; when the two coordinates constrain each other, or when the options are alternatives rather than simultaneous data, the correct structure is a *sum* (a disjoint union) or a subobject of a product, and the counting rule changes from $\cdot$ to $+$ or to something smaller.



$3 \cdot 2 = 6$ shirts: a product is a rectangle of options

3 Products in the two *Tárgytematika* documents

3.1 Kalkulus I (MBLK37E)

The syllabus text lists: *A valós számtest. Teljes indukció. Nevezetes egyenlőtlenségek. Polinomok, racionális törtfüggvények, gyökös, trigonometrikus, exponenciális függvények és inverzeik. Értelmezési tartomány, értékkészlet, inverz függvény, összetétel. Grafikonok vázolása... Függvények határértéke... műveletek... Elemi függvények deriváltja. Láncszabály...* The products hiding in it:

Syllabus item	Kind	Content, and the Lean name
A valós számtest — the field $\mathbb{R}$	multiplication map	The field operation is $m : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$. It is continuous (<code>mulMap_continuous</code>) and bilinear (<code>mulMap_bilinear</code>), but <i>not</i> linear (<code>mulMap_not_linear</code>).
Grafikonok vázolósa — graphs	product object	A graph lives in $\mathbb{R} \times \mathbb{R}$; a curve $t \mapsto (f(t), g(t))$ is continuous iff both coordinates are. <code>top_isProduct</code>
Értelmezési tartomány of $f \pm g, f \cdot g$	product object	The pairing $\langle f, g \rangle$ is defined on $\text{dom } f \cap \text{dom } g$: intersection is the product taken inside the poset of subsets.
Határérték ...műveletek	both	$\lim(fg) = \lim f \cdot \lim g$ is exactly “pair, then apply the continuous m ”. <code>tendsto_mul_via_product</code>
Elemi függvények deriváltja (Leibniz rule)	both	$(fg)' = f'g + fg'$ is the chain rule for $m \circ \langle f, g \rangle$, with the derivative of the bilinear m . <code>product_rule_via_product</code>
Összetétel, láncszabály	neither!	Composition is <i>not</i> a product construction; it is the categorical composition. Keeping $f \circ g$ and $f \cdot g$ apart is a graded skill. <code>comp_ne_pointwise_mul</code>
Teljes indukció, e.g. $n! < \left(\frac{n+1}{2}\right)^n$	multiplication map	Iterated products; induction is how a statement about an n -fold product is reduced to a binary one.
Exponenciális függvények és inverzeik	multiplication map	$\exp$ converts $+$ into $\cdot$, $\log$ converts $\cdot$ into $+$: a homomorphism between the two multiplication maps. <code>log_mulMap</code>

3.2 Lineáris algebra I (MBLK15E)

The syllabus text lists, among others: *Komplex számok... Vektorok és pontok a valós elem- n -esek körében, vektorok összege és skalárral való szorzása... Belső szorzat, vektorok hossza, háromszög-egyenlőtlenség, Cauchy–Schwarz-egyenlőtlenség... Mátrixműveletek és azok algebrai szabályai... Mátrixegyenletek, mátrixok inverze... Determinánsok... determinánsok szorzástétele, nemelfajuló mátrixok, Cramer-szabály.*

Syllabus item	Kind	Content, and the Lean name
valós elem- n -esek $\mathbb{R}^n$	product object	$\mathbb{R}^n$ is the n -fold product of $\mathbb{R}$: a linear map into $\mathbb{R}^n$ is exactly its n coordinate functionals, uniquely. <code>pi_isProduct</code>
Direct product $M \times N$ of spaces	product object	The product in the category of real vector spaces; a linear map into $M \times N$ is a pair of linear maps. <code>module_isProduct</code>
skalárral való szorzás $\mathbb{R} \times V \rightarrow V$	multiplication map	Bilinear, defined on a product of two <i>different</i> kinds of object.
Belső szorzat $\langle u, v \rangle$	multiplication map	$\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, built coordinatewise out of the same m . <code>inner_eq_sum_mulMap</code>
Mátrixműveletek: MN	multiplication map	Matrix multiplication is composition of linear maps: a map $\text{Mat} \times \text{Mat} \rightarrow \text{Mat}$. <code>toLin_matrix_mul</code>
determinánsok szorzástétele	multiplication map	$\det(MN) = \det M \cdot \det N$: $\det$ transports one multiplication map to the other. <code>det_mul_map</code>

Syllabus item	Kind	Content, and the Lean name
<i>mátrixok inverze, Cramer-szabály</i>	multiplication map	Inverses are about the multiplication map, not the product object; the categorical treatment is in <code>InverseOverlap.lean</code> .
<i>Egyenesek, síkok, hipersíkok</i>	product object	A plane is a product of two lines; a linear system is read off coordinatewise, i.e. through the projections of $\mathbb{R}^n$.
<i>Komplex számok $a + bi$</i>	both	$\mathbb{C} \cong \mathbb{R} \times \mathbb{R}$ as a product object (real and imaginary part), while complex multiplication is an extra bilinear map on it — the same object carrying both structures.

3.3 The overlap, in one sentence

Both courses use the *same* product object $\mathbb{R} \times \mathbb{R}$ and the *same* multiplication map m on it; the two courses differ only in which property of m they exploit:

One object $\mathbb{R} \times \mathbb{R}$, one arrow $m : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, two courses:		
<i>Kalkulus I</i>	m is continuous	$\implies \lim(fg) = \lim f \cdot \lim g, (fg)' = f'g + fg'$
<i>Lineáris algebra I</i>	m is bilinear	$\implies \langle u, v \rangle, MN, \det(MN) = \det M \det N$

`mulMap_continuous_and_bilinear` records both at once. And the reason $(fg)' \neq f'g'$ — the most common exam error — is precisely `mulMap_not_linear`: m is bilinear but *not* linear, so it does not commute with taking derivatives coordinatewise.

4 Products and multiplication maps in `kalkulus_gyakorlo.pdf`

Exercise	What it asks	Product structure to use
1.4 d)	Bound $\{p/q : p \in \{1, 2, 3, 4\}, q \in \{101, \dots, 104\}\}$	The index set is a product of two 4-element sets, so at most 16 values; the bounds are min / max over a rectangle. <code>exercise_1_4d_card_le</code>
1.12	Inequalities such as $\frac{(x-3)(x+2)}{(x-4)(x-8)} < 0$	A sign line is the statement that the sign map $\{\pm\} \times \{\pm\} \rightarrow \{\pm\}$ is the multiplication map of signs; solve factor by factor and multiply.
1.13	$ a_1 + \dots + a_n \leq a_1 + \dots + a_n $	$ xy = x y $: the absolute value is a homomorphism for m (proved in <code>TriangleOverlap.lean</code>).
1.14 c)	$\sum_i q^i = \frac{q^{n+1}-1}{q-1}$	Telescoping after multiplying by $(q-1)$: the trick is to <i>apply</i> a multiplication map to the whole identity.
1.15 a), b)	$\sum i(3i+1), \sum i(i+1)$	Each summand is a value of m on a pair; expand by bilinearity, then use $\sum i, \sum i^2$.
1.19 b)	Binomial theorem	$(a+b)^n$ is an n -fold product; expanding it is choosing, for each factor, a or b — a product of n two-element sets, so 2^n terms.

Exercise	What it asks	Product structure to use
2.10, 2.12–2.15	$f \circ g$, tables, equations	<i>Composition</i> , not product. The exercise trains exactly the distinction of Section 1. <code>comp_ne_pointwise_mul</code>
2.13 a)	$f(x) = x \sin x$	$m \circ \langle \text{id}, \sin \rangle$; differentiate with the Leibniz rule. <code>exercise_2_13a_deriv</code>
2.17 a), b)	Are $f + g$, $f \cdot g$ injective?	No in both cases: m itself is not injective, and no hypothesis on the factors repairs that. Counterexamples $\text{id} \cdot \text{id} = x^2$ and $\text{id} + (-\text{id}) = 0$. <code>exercise_2_17b</code> , <code>exercise_2_17a</code>
2.18	$f \circ g$ injective if f, g are	Yes — composition preserves monomorphisms, products do not. Compare with 2.17 to see the difference between the two operations.
2.19	A degree- n polynomial has $\leq n$ roots	A polynomial is a product of linear factors over its roots; the multiplication map is zero-divisor-free in $\mathbb{R}$.
3.2–3.4	Limits of rational functions	Factor, i.e. write the expression as $m \circ \langle \cdot, \cdot \rangle$, and use the product rule for limits. <code>tendsto_mul_via_product</code>
3.7	$\lim \frac{\sin 3x}{x}$, $\lim \frac{\tan 2x}{x}$, ...	Rewrite as a product of a known limit and a bounded/convergent factor, then multiply the limits.
3.9	One-sided limits of products/quotients	The sign of the product is the product of the signs; the finite factor multiplies the infinite one.
3.10	$\lim(1 + \frac{a}{x})^x$	An exponent is an iterated product: take log to convert it into a product, then into a $\frac{0}{0}$ limit. <code>log_mulMap</code>
4.1–4.2	Differentiate $x^5 5^x$, $x \cos^3 x$, ...	Every one of these is $m \circ \langle f, g \rangle$; apply the Leibniz rule. <code>product_rule_via_product</code>
4.3	Logarithmic differentiation, e.g. $\left(\frac{(x+1)(x-1)}{(x-2)(x+3)}\right)^5$	log turns a product into a sum, differentiate the sum, multiply back. The whole method is “move along the homomorphism log”. <code>log_mulMap</code>
4.11 a)	Split 20 so that the product is maximal	Maximise m restricted to the line $x + y = 20$: the answer is $10 \cdot 10 = 100$. <code>exercise_4_11a</code> , <code>exercise_4_11a_attained</code>
4.12, 4.14, 4.15	Area/perimeter optimisation	Area is a multiplication map of two side lengths, constrained by a linear condition — the same shape of problem as 4.11.

5 Using the concept in a timed exam

5.1 A four-step protocol

Step 1. Name the szorzat. Ask: is this a *pair* (product object) or a *value* (multiplication map)? Write the answer in the margin: “ $\times$ ” or “ m ”. Ten seconds, and it selects the toolbox.

Step 2. Split. If it is a value $F = f \cdot g$, write $F = m \circ \langle f, g \rangle$ — in practice, box f and g

separately. If it is a pair, handle the two coordinates separately: everything about a map *into* a product is coordinatewise.

Step 3. Apply the property of m that the question needs.

- Limit? Use continuity of m : $\lim(fg) = \lim f \cdot \lim g$ — but only when both factors have finite limits.
- Derivative? Use bilinearity of m : $(fg)' = f'g + fg'$.
- Sign or inequality? Use that m preserves the order in each variable separately when the other factor is ≥ 0 .
- Equation = 0? Use that $\mathbb{R}$ has no zero divisors: $fg = 0 \iff f = 0$ or $g = 0$.
- Too many factors? Take log and work with a sum.

Step 4. Reassemble and check one number. Substitute a convenient value ($x = 0$, $x = 1$) into your answer and into the original. Ten seconds, and it catches the common slips.

5.2 Four worked answers, in exam format

(a) **Exercise 4.2, differentiate $x^5 5^x$.** Step 1: value, so m . Step 2: $f = x^5$, $g = 5^x$. Step 3: Leibniz,

$$(x^5 5^x)' = 5x^4 5^x + x^5 5^x \ln 5 = x^4 5^x (5 + x \ln 5).$$

Step 4: at $x = 0$ the function is 0 with a fifth-order zero, and the answer vanishes there. product_rule_via_product

(b) **Exercise 3.7 b), $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$.** Step 1: value. Step 2: $\frac{\sin 3x}{x} = \underbrace{\frac{\sin 3x}{3x}}_{\rightarrow 1} \cdot 3$. Step 3: both

factors have finite limits, so multiply them: the limit is 3. Step 4: for $x = 0.01$ the quotient is ≈ 2.9996 . tendsto_mul_via_product

(c) **Exercise 2.17 b), is $f \cdot g$ injective for injective f, g ?** Step 1: value. Step 2: the composite $m \circ \langle f, g \rangle$ can only be injective if m separates the pairs that $\langle f, g \rangle$ produces — and m is very far from injective. Step 3: exhibit one counterexample, which is all an exam needs: $f = g = \text{id}$ gives x^2 , and $1 \neq -1$ while $1^2 = (-1)^2$. Answer: **no**. Compare 2.18: for $f \circ g$ the answer is *yes*, because composition, unlike m , preserves injectivity. exercise_2_17b

(d) **Exercise 4.11 a), split 20 so that the product is maximal.** Step 1: it is a multiplication map restricted to a line — a map out of a product, constrained. Step 2: eliminate one coordinate, $y = 20 - x$, i.e. parametrise the line by a single variable; the object to maximise is $P(x) = x(20 - x)$. Step 3: $P'(x) = 20 - 2x = 0$ gives $x = 10$, and $P'' = -2 < 0$, so $x = y = 10$, $P = 100$. Step 4: $9 \cdot 11 = 99 < 100$. A one-line alternative without calculus: $xy = 100 - (x - 10)^2 \leq 100$. exercise_4_11a

5.3 A time budget for a 90-minute paper

Minutes	Phase	What products buy you
0–5	Triage	Mark every “.” in the paper as $\times$ (pair) or m (value). The m -items are usually the fastest marks.
5–30	Routine differentiation/limits	Leibniz and the product limit law, mechanically; do not re-derive them.
30–60	Multi-step items	Logarithmic differentiation, factorised sign analysis, optimisation of an area.
60–80	Conceptual items	Injectivity/surjectivity of $f \cdot g$ versus $f \circ g$; counting arguments; “is this set a product?”
80–90	Checks	Substitute one value in each answer; verify each “product” claim by the test of Definition 1.1.

5.4 Five traps that cost marks

1. $(fg)' = f'g'$. False; m is bilinear, not linear (`mulMap_not_linear`). The correct rule has two terms.
2. $\lim(fg) = \lim f \cdot \lim g$ **applied when one factor has no finite limit**. The hypothesis of `tendsto_mul_via_product` is that *both* limits exist and are finite. For $0 \cdot \infty$ rewrite first.
3. **Confusing $f \circ g$ with $f \cdot g$** . Different operations with different preservation properties: composition preserves injectivity (2.18), pointwise product does not (2.17 b).
4. **Assuming a constrained set of pairs is a product**. “All (x, y) with $x + y = 20$ ” is a subobject of $\mathbb{R} \times \mathbb{R}$, not a product — that is exactly why 4.11 is an optimisation problem and not a two-line calculation.
5. **Adding when the structure multiplies (or the reverse)**. Independent choices multiply (`card_prod_eq_mul`); alternatives add. Deciding which one applies is Step 1 of the protocol.

5.5 The one-line summary to memorise

A *product object* is a pair: maps *into* it are handled coordinatewise.
A *multiplication map* is an arrow out of a pair: it is continuous (so limits pass through) and bilinear (so derivatives obey Leibniz), but it is neither linear nor injective.